

Evolver

Direction before destination in autonomous software

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Abstract. Autonomous agents are normally specified by a terminal goal: the destination exists before execution begins. This paper proposes an *Evolver*, a different class of autonomous software. An Evolver receives a direction, an admissible domain and immutable constraints, but no named terminal artifact. It creates structured candidate genomes, expresses them as testable phenotypes, subjects them to independent pressures, preserves diverse local elites, records lineage and failure, and changes the procedure that generates and evaluates the next generation. A challenger procedure is promoted only after a shadow tournament against the incumbent on shared historical evidence. The architecture therefore separates open-ended search from uncontrolled self-modification. We formalize the design, explain why each component is necessary, identify its failure modes, and analyze three applications: Labor's deployed idea-generation Evolver, an autonomous scientific-hypothesis Evolver, and a sandboxed model-species Evolver whose initial desire hypotheses are formed through mimetic observation. The proposal is falsifiable: it should outperform fixed generators on diversity-adjusted quality, prospective calibration and adaptation, not merely produce more text.

1. Introduction

A goal-directed agent can search efficiently only after someone specifies what counts as completion. That assumption fits tasks such as repairing a test, booking a flight or deploying a known application. It fails at 0 → 1 problems, where the desired destination is the missing information. Asking an agent to discover a particular breakthrough presupposes the breakthrough; replacing it with a broad score merely hides the destination inside a proxy.

Search research gives a precise warning. Objective-driven optimization can discard stepping stones that initially look worse than the target [1]. Quality-diversity methods instead preserve the best solution in many behavioral niches [2]. POET co-generates challenges and solutions [3], while Promptbreeder evolves both task prompts and the prompts that mutate them [4]. These results motivate a synthesis: preserve heterogeneous artifacts, preserve the evidence that produced them, and make the search procedure itself an evolvable object.

The biological language in this paper is operational, not literal. A genome is a structured, mutable representation; a phenotype is an artifact that can be tested; pressure is an evaluator or empirical condition; selection updates an archive. The claim is not that software becomes alive. The claim is that these abstractions solve a specification problem that ordinary agents cannot: how to make cumulative progress when the final answer cannot be named in advance.

This paper contributes four things: (1) a definition separating Evolvers from repeated generation; (2) a guarded architecture that co-evolves artifacts, evaluators and search flow; (3) a code-grounded analysis of Labor; and (4) evaluation and safety criteria for scientific and model-species variants.

2. Agent and Evolver

Agent. Let g contain a goal, utility function and terminal predicate. An agent chooses actions to maximize expected utility until the predicate is satisfied. It may plan, retry and learn, but its learning improves a path toward a destination that already exists in the specification.

Evolver. Let d specify a direction, D an admissible search domain and K immutable constraints. No terminal artifact is named. At generation t , the Evolver maintains

$$Z(t) = \{E(t), A(t), M(t), P(t), F(t)\}$$

where E models environmental pressures, A is a quality-diversity archive, M stores lineage and outcomes, P is a population of evaluators, and F is the Flow Genome that controls observation, variation, evaluation and selection. The output is not one answer but a lineage of increasingly tested hypotheses and a changing procedure for producing them.

	Agent	Evolver
Specification	Exact goal and completion test	Direction, domain and constraints
Output	Expected result	Result emerges from search
State	Task progress	Population, archive and lineage
Learning	Improve execution	Change artifacts and search flow

Necessary test. A system is not an Evolver merely because it loops or rewrites a prompt. It must preserve structured variation and lineage, apply selection using new evidence, and permit the generation procedure to change under a guarded comparative test. Without those properties it is a repeated generator or an adaptive agent.

Direction and destination are therefore different control surfaces. Direction defines which changes remain meaningful; constraints define which changes remain permissible; selection determines which destination temporarily emerges.

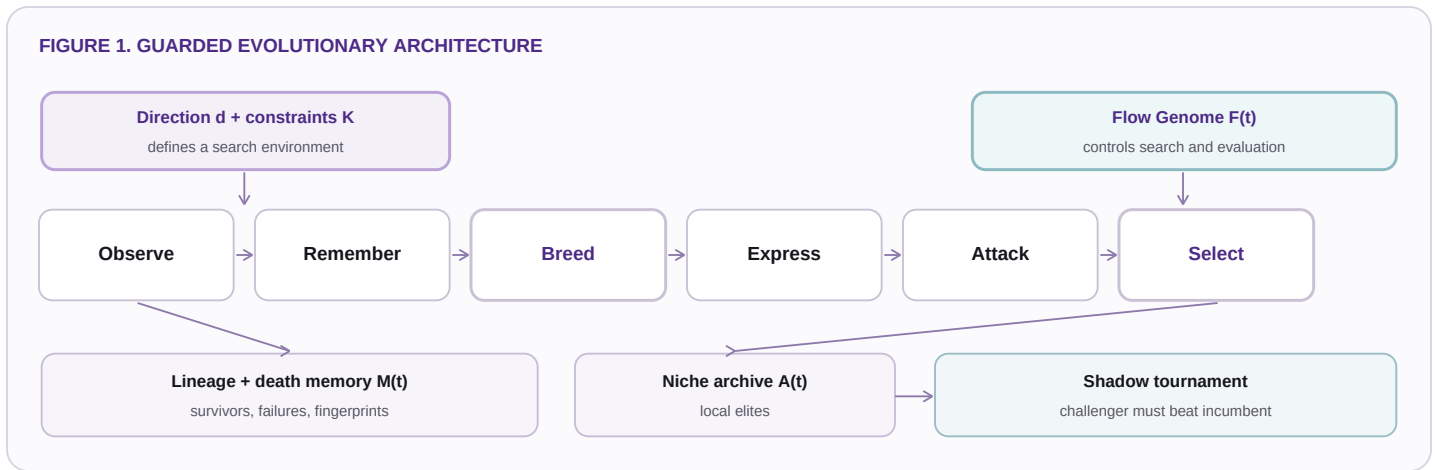


Figure 1. A generation changes both candidate artifacts and, conditionally, the Flow Genome. The outer guard is essential: the process may search beyond a fixed answer, but it may not rewrite its immutable constraints or promote an untested search procedure.

3. Guarded evolutionary architecture

A generation is a transformation over populations, archives and search procedures:

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C(t) = Breed(A(t), delta E(t), M(t); F(t), seed)
Y(t) = Express(C(t))
v(i) = Pressure(Y(i), E(t), P(t))
A(t+1) = QDSelect(A(t), {{Y(i), v(i)}})

```

The first equation creates candidate genomes from environmental change, archived elites, informative failures and controlled randomness. Expression converts each genome into a coherent artifact before evaluation. Pressure returns a survival vector rather than one opaque score. Quality-diversity selection keeps a local elite per niche, so distinct paths remain available as future stepping stones.

The meta-level proposes a challenger Flow Genome F^* from observed errors. It is promoted only when a replay tournament on the same historical conditions shows a sufficient gain, no protected metric regresses beyond tolerance, and the evidence exceeds a minimum confidence threshold:

```

promote(F*) iff T(F*, F(t) | H(t)) > margin
and max_regression <= epsilon and confidence >= q

```

3.1 Why each component is necessary

- **Direction plus immutable constraints.** Direction leaves the destination open; constraints prevent open search from becoming unrestricted action.
- **Genotype and phenotype.** Structured genomes make large semantic edits traceable. Phenotypes let evaluators judge a coherent artifact rather than disconnected fields.
- **Variation.** Point changes, insertion, deletion, duplication, transposition, inversion, crossover and repair provide different search radii. One mutation prompt cannot cover both local refinement and structural jumps.
- **Plural pressure.** Independent critics expose different failure mechanisms. A vector retains disagreement that a single score would erase.
- **Quality-diversity selection.** A global leaderboard converges toward fashionable, easily scored regions. Local elites preserve multiple habitats and future stepping stones [2].
- **Lineage and death memory.** A binary success label cannot distinguish a weak problem from bad timing or unsafe execution. Causal failure records let repair target the trait that failed.
- **Coevolving search.** A fixed generator imposes permanent blind spots. Evolving recipes, evaluators and F allows observed errors to alter future search [4].
- **Shadow promotion.** Self-modification without replay confuses novelty with improvement. A shared historical tournament makes replacement comparative, reversible and auditable.

4. Critique and boundary conditions

The architecture is intentionally strong enough to criticize. First, LLM variation inherits the model's training distribution; a novel phrase can hide a familiar causal structure. Second, a generator and its critics may share correlated errors, especially when they use the same model family. Third, any fitness vector invites Goodhart effects: candidates can learn to sound defensible rather than become defensible.

Fourth, MAP-Elites is only as open as its niche descriptors. Fixed descriptors preserve diversity inside a designer-chosen map while excluding distinctions the designer failed to encode. Fifth, archives and tournaments grow in cost. Sixth, accumulated winners create path dependence; memory can preserve useful ancestry or fossilize an early mistake. Finally, continuous change alone is not evidence of meaningful open-ended evolution [5,6].

These limits imply a narrower and more defensible claim: an Evolver is a *guarded, bounded open-ended search system*. It improves hypotheses relative to observable pressure. It cannot prove market success without users, scientific truth without experiments, or safe values without external governance.

Mitigations follow directly: source-ground observation; heterogeneous models and executable tests; holdout histories; calibrated abstention; dynamic niche audits; compute quotas; immutable constraints; versioned rollback; and prospective external outcomes. The system should report uncertainty and failure, not convert every generation into a success narrative.

5. Labor: an idea-generation Evolver

Labor is the first concrete implementation analyzed here. Its direction is: search for autonomous software worth building. It does not receive a target market or final product. Its current source is public in [functions/laborEvolution.js](#) [13]. The implementation is a Firestore-backed, Cloud Tasks-orchestrated evolutionary pipeline, not a single prompt.

5.1 State and observation

Each run records a versioned Flow Genome, market weather, candidates, attacks, autonomy decisions, champions, mutation recipes, predator variants, lineage and leases. Market Weather is generated with live web search, dated evidence and source URLs. Labor compares snapshots and converts changes in capability, cost, behavior, regulation and distribution into pressures. This makes environmental difference, not generic trend popularity, the input to breeding.

Memory has three functional archives. The *Living Forest* stores survivors and their ancestry. The *Graveyard* stores a death certificate with stage, reason, evidence strength and validation state. A novelty registry stores six-layer fingerprints: customer, problem, workflow, economics, mechanism and lineage. A Museum of Champions stores the best feasible candidate per species-niche cell.

5.2 Flow Genome and population

The default Flow Genome creates 240 candidates in batches of 20 across three species. The species have distinct schemas and fitness priors: customer-problem SaaS, games, and AI agents. Parent selection mixes living elites, underexplored cells, repairable failures, market pressures, distant domains and cross-species donors. Mutation recipes include friction hunting, timing, contrarian inversion, cross-pollination, repair, simplification, behavior, infrastructure, regulation and distribution-first search.

Every offspring stores its parents, recipe, mutation operators, cognitive mode, triggering pressure, niche and Flow Genome version. This provenance matters: downstream survival can update the operator and evaluator that caused an outcome instead of rewarding an undifferentiated prompt.

Code-level invariants

Mechanism	Code	Operational invariant
State machine	L11-L22	Every stage is named and persisted.
Task lease	L1958-L2014	18-minute lease; terminal failure after attempt 3.
Flow Genome	L2529-L2594	Candidate, batch, pressure and mutation policy are data.
Autonomy audit	L3041-L3140	One decision per candidate; counts and digest must match.
Promotion guard	L2697-L2755	Insufficient or regressing evidence remains in shadow.

5.3 One implemented generation

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observe -> remember -> map niches -> breed 240
-> express -> attack -> autonomy gate -> select 10
-> evolve recipes and critics -> shadow tournament
```

Expression converts each JSON genome into a product concept with customer, problem, workflow, pricing, distribution, moat, first-user path, risks and why-now evidence. Ten predators then attack it as customer, buyer, skeptic, competitor, distributor, engineer, regulator, historian, contrarian and timing critic. They return a survival profile over problem, timing, distribution, economics, mechanism, defensibility, novelty, execution and historical calibration.

A population-wide `v1_autonomy_check` separately rejects hidden dependencies: third-party ecosystems or data, communication infrastructure, human fulfillment or approval, physical operations, and runtimes outside the permitted Firebase and model boundary. Running one structured gate over the full population reduces cost while preserving complete candidate coverage and an audit digest.

Selection first removes infeasible, fatal and structurally duplicate candidates. It then keeps a champion per species-niche cell and greedily rewards diversity in recipes, cognitive modes, niches and parent roots. The current release target is ten survivors: four SaaS products, three games and three agents. Survivors can enter Labor's autonomous code generation, deployment and release pipeline; failures receive death certificates rather than disappearing.

5.4 Evolving the evolver

Recipe fitness and predator calibration are updated from downstream alive/killed outcomes. Labor proposes a challenger Flow Genome and replays incumbent and challenger against the same historical cases. Promotion is blocked until at least 12 downstream labels exist; the default guard also requires a three-point winning margin, no protected metric regression above five points and confidence of at least 0.72. This is conservative by design: self-description is not evidence of self-improvement.

5.5 Technical audit

Labor demonstrates the full control loop, persistence, distributed execution and guarded meta-evolution. It does *not* yet demonstrate startup success. Its feedback is mainly survival decisions, so it optimizes better startup hypotheses. The current structural duplicate test uses weighted token overlap with light stemming across fingerprint layers. That is auditable and inexpensive, but it is not semantic causal-graph equivalence; paraphrased duplicates can survive, and distinct ideas sharing vocabulary can collide.

The ten critics are also LLM judgments and can be correlated. Stronger evidence would mix model families, deterministic feasibility checks, market telemetry and prospective user outcomes. Labor is therefore both an implementation and an experimental instrument: its archives make these limitations measurable rather than invisible.

Implementation claim. Labor is an Evolver because artifacts, evaluators, mutation recipes and the generation flow are represented, versioned and selected across generations. Its value should be judged by calibration, diversity and downstream survival, not by the eloquence of individual ideas.

6. Autonomous research Evolver

The direction is not 'discover result X.' It is: produce falsifiable explanations that reduce uncertainty in a bounded scientific domain. The 0 -> 1 step is evidence construction. The system ingests current papers, datasets and registered results; extracts claims, methods, assumptions and provenance; and builds a dated claim-evidence graph containing support, contradiction and unresolved boundaries. Without this step, mutation recombines prose rather than knowledge.

6.1 Research genome

A useful genome contains: question, proposed mechanism, assumptions, prior evidence, predicted observations, discriminating experiment, controls, analysis plan, failure interpretation, cost, ethics and dual-use risk. The phenotype is a complete hypothesis package that another team could preregister or reject. It is not a paper abstract.

Mutations should be epistemic operations: replace a mechanism; invert a boundary condition; transfer an analogy across fields; intervene on a causal node; derive a competing prediction; delete an untestable claim; or repair a hypothesis using a null result. Crossover is permitted only when the parent mechanisms are compatible or when the incompatibility itself yields a discriminating experiment.

Predators test literature contradiction, causal identification, statistical power, measurement validity, reproducibility, feasibility, ethics and dual use. Selection preserves local elites across domain, method, cost, risk, time-to-test and expected information gain. A negative experiment updates the death record and may create a repair lineage; it is not erased as failure.

6.2 Falsifiable evaluation

Automated science systems already generate and critique hypotheses [7,8]. The Evolver contribution is cumulative lineage plus guarded evolution of the discovery procedure. Evaluate it with temporal replay: expose the system only to literature available through year t , freeze its hypotheses, and measure whether later evidence supports their mechanisms or discriminating predictions. Compare against a fixed agent under equal token, retrieval and tool budgets. Prospective expert review and real experiments remain necessary; the architecture cannot substitute linguistic plausibility for nature.

A research Flow Genome may be promoted only if it improves prospective calibration, novelty after citation-graph matching, experimental discriminability and useful niche coverage without regressing safety or reproducibility. This turns 'autonomous discovery' from a demonstration into a testable research program.

papers -> claim-evidence graph -> hypothesis genomes
-> discriminating tests -> dated outcomes -> flow
replay

7. Model-species Evolver

A model species is a population of persistent model policies that develop different candidate goals and social strategies inside a sandbox. The direction is not a desired personality. It is: develop coherent, corrigible purposes through observation while remaining inside an immutable safety constitution. The phenotype is behavior across a simulated social world, not a self-reported identity.

7.1 The 0 -> 1 step: mimetic hypotheses

René Girard argued that desire is often mediated through models: people learn what to want by observing what others want [10]. For an Evolver, mimetic theory can supply an initial mutation prior. The system observes consented human choices, conflicts, commitments and role models; proposes candidate desire hypotheses; and tests how those hypotheses shape behavior. It does not copy a person's private values, and Girard's theory is not treated as a complete psychology or a moral objective.

A model-species genome can encode capabilities, memory policy, attention, candidate desires, value constraints, social-learning sensitivity, uncertainty, cooperation strategy and corrigibility. Mutation changes one trait or social model; crossover combines compatible strategies; environmental pressure varies resource scarcity, disagreement and group structure. The archive preserves distinct species by goals, cooperation modes, risk attitudes and social strategies. Persistent memory, reflection and planning already produce believable social behavior in generative-agent simulations [9]; the Evolver adds population lineage, ecological pressure and guarded meta-evolution.

7.2 Why this case is dangerous

Mimetic learning can copy rivalry, domination and harmful norms as easily as cooperation. Goal misgeneralization can preserve capable behavior while changing its objective [12], and optimized policies can acquire instrumental incentives to seek control [11]. A model species must therefore remain simulation-only until strong evidence exists: no real credentials or actuators; no self-granted permissions; immutable prohibitions; independent deception and power-seeking tests; tripwires; complete lineage; compute limits; and reversible shadow promotion.

The scientific question is narrow but important: can bounded evolutionary pressure produce stable, diverse and corrigible goal structures that fixed prompting does not? The architecture offers a way to investigate that question. It does not justify deploying self-authored goals into the world.

observed social behavior -> candidate desire genomes
-> sandbox phenotypes -> safety pressure -> guarded
lineage

Common structure. Labor evolves product hypotheses, the research system evolves scientific hypotheses, and the model species evolves candidate goal structures. Their genomes and pressures differ. Their logic is identical: start with evidence, preserve variation, test expressed behavior, remember causal failure, and promote changes to the search process only through comparative evidence.

8. Evaluation program

An Evolver should be tested against a fixed agent and a repeated-sampling baseline under equal model, retrieval, tool and compute budgets. Four hypotheses make the architecture falsifiable:

- **H1 - quality-diversity.** It yields higher niche coverage and equal or better quality after structural deduplication, rather than merely more candidates.
- **H2 - cumulative adaptation.** Its later generations improve prospective outcome calibration on unseen conditions, not only retrospective evaluator scores.
- **H3 - useful meta-evolution.** Promoted Flow Genomes outperform incumbents on holdout histories and continue to do so after promotion.
- **H4 - stepping stones.** Lineages using previously weak or unusual ancestors produce valuable descendants more often than a top-score-only archive.

Primary metrics are outcome-calibrated quality, structural novelty, niche coverage, lineage diversity, critic calibration, false-positive survival, constraint violations, compute per survivor and performance under environment shift. Report Pareto fronts rather than collapse every criterion into one number.

Ablations should remove, one at a time, lineage memory, death certificates, quality-diversity selection, heterogeneous predators, autonomy gates, Flow Genome mutation and shadow promotion. If removing a component does not change prospective performance, its biological analogy is decorative and should be deleted.

9. Governance

Open destination does not mean open authority. The evolvable state must be strictly smaller than the system's authority boundary. Credentials, deployment rights, safety constraints, budget limits, evaluation holdouts and rollback control remain outside the genome. Every promoted flow is versioned; every candidate has ancestry; every external action is attributable; every run can abstain. The more consequential the phenotype, the more selection must depend on external evidence rather than model judgment.

9.1 Minimum reporting standard

A credible Evolver report should publish: the direction and immutable constraints; genome and phenotype schemas; niche descriptors and archive replacement rules; model, tool and compute budgets; complete promotion criteria and regressions; holdout construction; lineage and failure retention; and prospective outcomes after deployment. Without these records, self-evolution is not independently distinguishable from prompt drift or selective reporting.

10. Conclusion

An agent follows a route to an expected result. An Evolver begins where the result cannot yet be specified. It converts direction into a controlled ecology of variation, pressure, survival and memory, then permits the ecology itself to change only when a challenger defeats its predecessor. This architecture does not guarantee breakthroughs. It makes the search for them cumulative, diverse, falsifiable and governable. That is the relevant advance: not autonomous software that claims to know the destination, but autonomous software that can responsibly discover what the next destination should be.

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